

Java Notes - Class, Objects, and Methods

1. Class in Java

Definition: A class is a user-defined blueprint or template used to create objects.

Everything in Java is written inside a class.

A class contains variables (properties) and methods (behavior).

A class itself does not occupy memory; memory is allocated only when an object is created.

Syntax:

```
class ClassName {  
    // variables  
    // methods  
}
```

2. Object in Java

An object is an instance of a class. Every object has:

- Properties (variables/data)
- Methods (behavior/actions)

Objects occupy memory.

3. Methods in Java

A method is a block of code that performs a specific task.

Syntax:

```
public int add() {  
    return 10 + 20;  
}
```

Parts: Access Modifier, Return Type, Method Name, Parameters.

4. Creating an Object

Syntax:

```
ClassName ref = new ClassName();
```

Example:

```
Student s = new Student();
```

The **new** keyword creates an object and allocates memory.

5. Calling a Method

Use the reference variable and dot operator.

```
s.display();
```

6. Methods with Parameters

Example:

```
public void greet(String name) {  
    System.out.println("Hello " + name);  
}
```

Call: `s.greet("Ritwik");`

7. Method Returning a Value

Example:

```
public int add(int a, int b) {  
    return a + b;  
}
```

Quick Revision

- Class = Blueprint
- Object = Instance of a class
- Class does not occupy memory
- Object occupies memory
- Method performs an action
- new creates an object
- Reference variable stores the object's reference
- Parameters receive values
- Return type specifies what the method returns